

Homework 2
COMS 3310
Due Wednesday Feb 18th at 11:59pm

Late Submission: Thursday Feb 19th, 11:59PM (15% Penalty)

Please read the HW guidelines (see next page). Failure to follow the guidelines will result in a lower grade.

Notation: $\mathbb{N}$ is the set of natural numbers, i.e.,

$$\mathbb{N} = \{0, 1, 2, \dots\}.$$

1. Let $\Sigma = \{a, b, c\}$. Let $|x|_a$ denote the number of times the symbol a occurs in the string x . For example, $|aabab|_a = 3$, $|aabab|_b = 2$ and $|aabab|_c = 0$.

Corresponding to every natural number $k \geq 2$, we can define a language

$$A_k = \{x \in \Sigma^* \mid |x|_a \% k = 1\}$$

- (a) Draw the state diagram of a DFA for the language A_4 .
 - (b) Generalize it by defining a DFA M_k for the language A_k using math notation only. For this subproblem do not draw a state diagram. Instead provide a specification of the parameterized DFA as a 5-tuple. Parameterized means that for any fixed k , the specification will give you a different DFA. There is no need to prove the correctness of the specification, just write a correct specification.
2. Construct a DFA for each of the following languages by drawing a state diagram.
 - (a) $\Sigma = \{a, b\}$. $L_1 = \{x \in \Sigma^* \mid x \text{ does not contain the substring } aba\}$. Hint: Start by sketching a DFA for the language $\{x \in \Sigma^* \mid x \text{ contains the substring } aba\}$.
 - (b) $\Sigma = \{a, b\}$. $L_2 = \{x \in \Sigma^* \mid x \text{ ends with } ab\}$.
 3. Construct a DFA for each of the following languages by drawing a state diagram.
 - (a) $\Sigma = \{a, b\}$. L_3 contains strings that start with a and have even length, or start with b with a length which is a multiple of three.
 - (b) $\Sigma = \{a, b, c\}$. $L_4 = \{x \in \Sigma^* \mid x \text{ contains the substring } aabaac\}$.

4. Prove that the following language is not regular. Your proof must apply the Pigeonhole principle directly, as we did in lectures. You should not use concepts such as the pumping lemma, the Myhill-Nerode theorem or periodicity. Let $\Sigma = \{a, b, c\}$.

$$L_5 = \{a^n b c^{2n} \in \Sigma^* \mid n \geq 0\}.$$

5. Prove that the following language is not regular. Your proof must apply the Pigeonhole principle directly, as we did in lectures. You should not use concepts such as the pumping lemma, the Myhill-Nerode theorem or periodicity. Let $\Sigma = \{a\}$.

$$L_6 = \{x \in \Sigma^* \mid |x| \text{ is a power of two}\}.$$

By this I mean natural number powers only, so $L_6 = \{a, a^2, a^4, a^8, \dots\}$

Guidelines:

- Please submit your HW via Gradescope.
- You must handwrite your solutions.
- If your handwriting is not legible, it will not be graded.
- Start the answer to each question on a separate page.
- You are allowed to collaborate with *others*, while doing your homework.
- Definition of *others*: Students who are taking the course this semester.
- You can consult GenAI tools.
- However, you should write the final solutions alone, without consulting anyone. Your writing should demonstrate that you understand the proofs completely.
- For each problem, you should acknowledge all the individuals you consulted and whether you used any GenAI resources. This will not affect your grade. Failure to acknowledge is considered *academic dishonesty*, and it will affect your grade.
- When proofs are required, you should make them both clear and rigorous. Do not hand wave.
- Any regrading requests should be made within one of returning the homework. No requests will be entertained after this one week period.